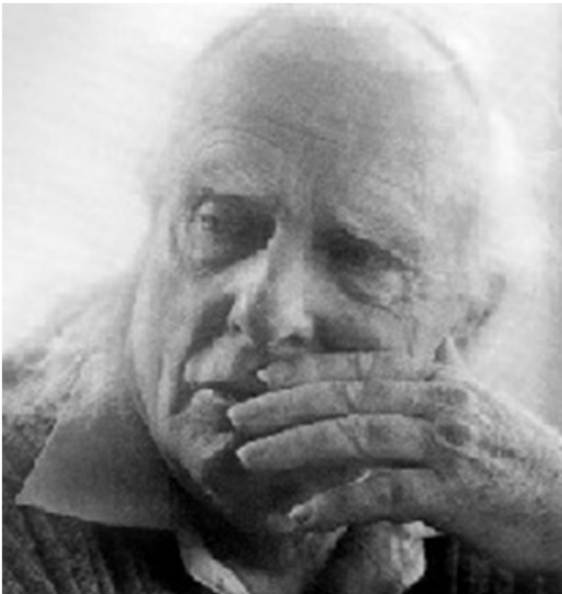


Logic and Conversation (Grice, 1975)

Siyuan

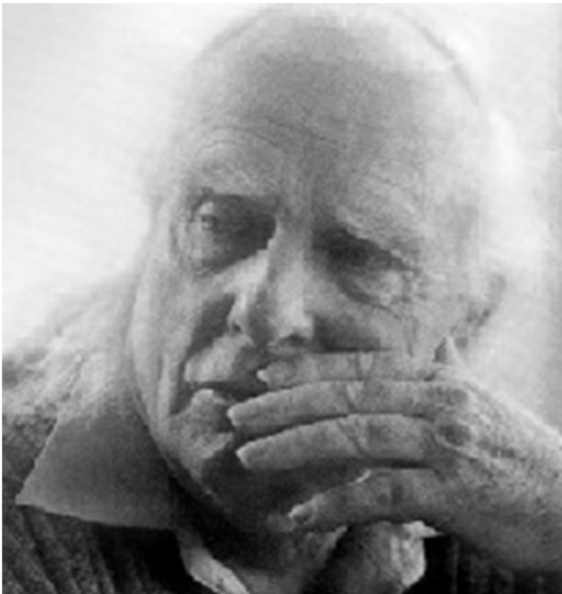
Grice (1975)



Implicature: what the speaker IMPLY

Paul Grice (1913-
1988)

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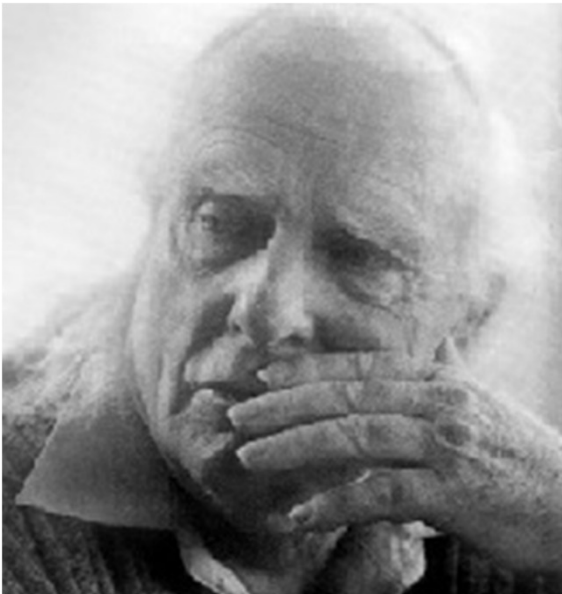


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Implicature: what the speaker IMPLY
conventional: without knowing the
circumstances of the utterance, the
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implicature

- *He is in the grip of a vice*

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- *He is in the grip of a vice*

Conversational Implicature:

Unconventional; happen in certain situations

- *What do you think of the upcoming presentation?*
- *Well, I'm still alive!*

Cooperative Principles / Gricean maxims

QUANTITY: Make your contribution as informative as is required;
and do not make it more informative as is required.

Cooperative Principles / Gricean maxims

Quality: Do not say what you believe to be false,

and do not say that for which you lack adequate knowledge.

Cooperative Principles / Gricean maxims

Relation: Simply, be relevant.

Cooperative Principles / Gricean maxims

MANNER:

- Avoid obscurity of expression
- Avoid ambiguity
- Be brief
- Be orderly

Failure to fulfil a maxim

- Lying & misleading
- Opt out conversation
- Not able to be informative enough with enough evidence
- Maxim Exploited: generate **conversational implicature**

Pragmatic Language Interpretation as Probabilistic Inference (Goodman & Frank, 2016)

Siyuan

Games



Games



My friend has a hat.

Games



My friend has a hat.

My friend has glasses.

A simple RSA model

Assumption 1: The speaker S is an utility-maximizing agent.
The listener L infer the state of the world w when receiving and utterance u

$$P_L(w|u) \propto P_S(u|w)P(w).$$

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Assume the listener is simple (L_{lit}), then the utility can be represented (in a simple way) as:

$$U(u; w) = \log P_{Lit}(w|u).$$

$$P_{Lit}(w|u) \propto \delta_{\llbracket u \rrbracket}(w)P(w).$$

Let's go back to the game...

My friend has glasses.



For a Literal Listener:

My friend has glasses -> G or HG

My friend has a hat -> HG

Let's go back to the game...

My friend has glasses.

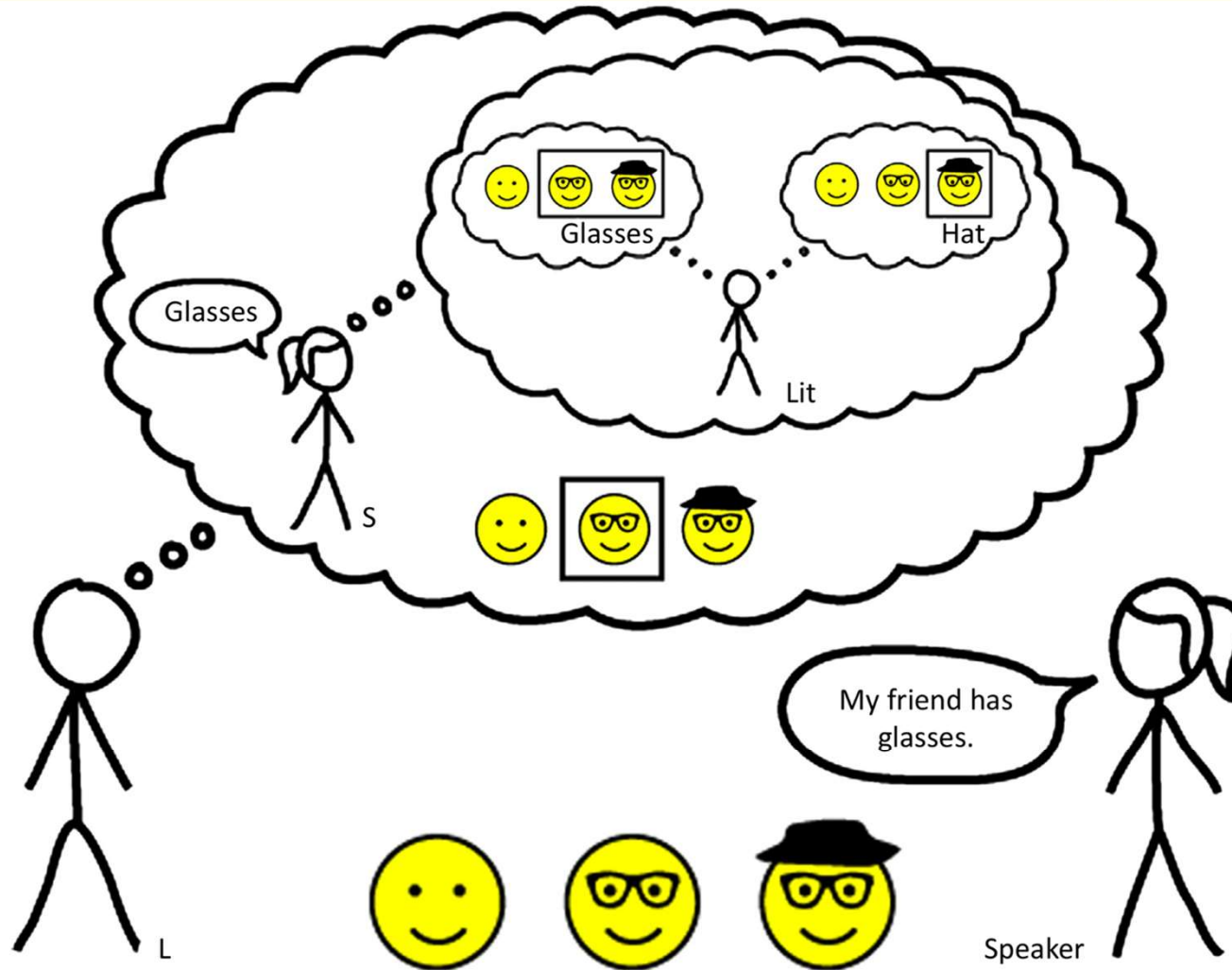


For a Literal Listener:

My friend has glasses -> G or HG

My friend has a hat -> HG

For a rational speaker: if communicating HG, should say 'my friend has a hat'



More complex versions of RSA

Different Levels of Recursion:

S_1 reasons on literal listener (L_{lit} or L_0 L_1); reasons on S_1 S_2 ,
 reasons on S_1 ... (different groups of people have different levels)

Elaborated Versions of Speaker Utility:

Cost: $U(u; w) = \log P_{Lit}(w|u) + \text{cost}(u)$

Speaker Uncertainty: $U(u; k) = \mathbb{E}_{P(w|k)}[U(u; w)],$

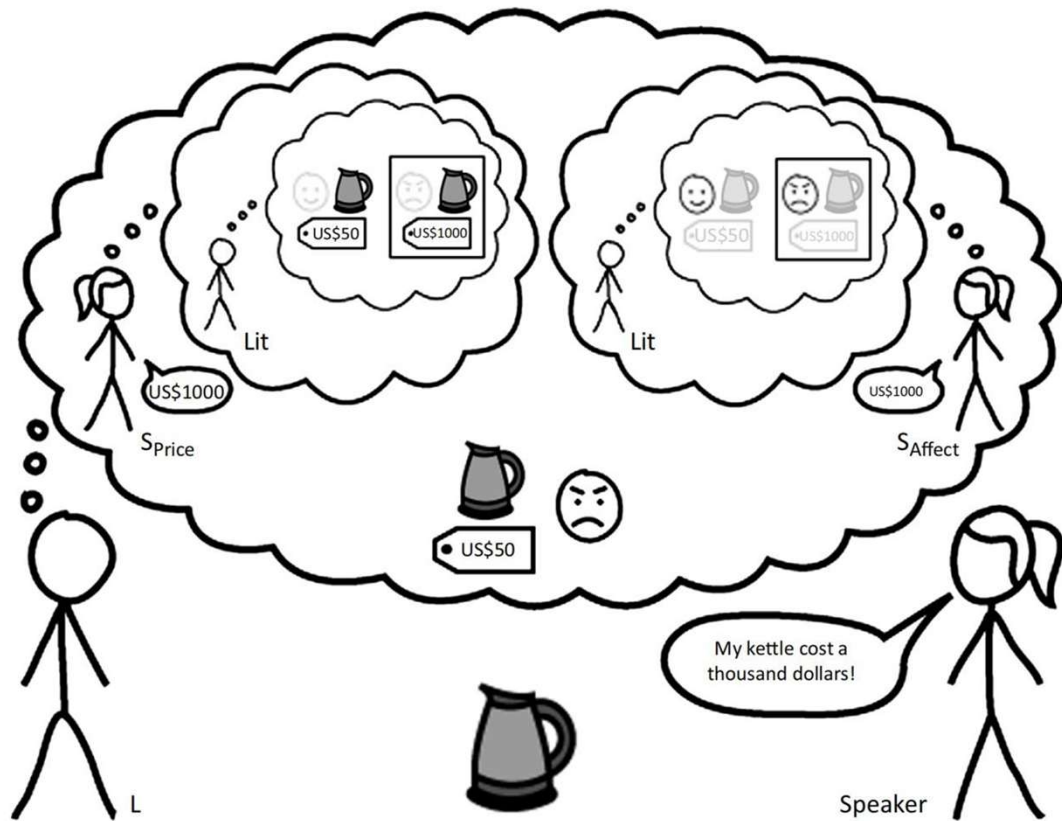
Topic relevance: $U(u; w, t) = \log \sum_{w' \text{ s.t. } t(w')=t(w)} P_{Lit}(w'|u).$

Uncertainty RSA (uRSA)

Joint inference: **what kind of speaker?** what kind of world?

$$P_L(w, s|u) \propto P_S(u|w, s)P(s)P(w)$$

Case study: hyperbolic utterances



Related work

Frank, Michael C., and Noah D. Goodman. "Predicting pragmatic reasoning in language games." *Science* 336.6084 (2012): 998-998.

Monroe, Will, and Christopher Potts. "Learning in the Rational Speech Acts Model." *Proceedings of the Amsterdam Colloquium*. 2015.

Unified Pragmatic Models for Generating and Following Instructions (Fried et al, 2017)

Siyuan

Motivation

- A pragmatic speaker might be able to reduce ambiguity when **generating instructions** by reasoning about the listener.
- A pragmatic listener might be able to overcome ambiguity when **following instructions** by reasoning about the speaker.
- Models of **pragmatic reasoning** has been largely restricted to simple reference games - can they be applied to complex tasks?

Task formulation: instruction following

- input:
 - a sequence of direction sentences: $d_1, d_2, \dots, d_K$
 - At each time t , the current world state y_t
- action:
 - At each time t , choose an action a_t
- goal:
 - reach the correct final state

Task formulation: instruction generation

- input:
 - a sequence of actions: $a_1, a_2, \dots, a_T$
 - the world state at each action: $y_1, y_2, \dots, y_T$
- task:
 - generate a sequence of direction sentences: $d_1, d_2, \dots, d_K$
- goal:
 - enable human listeners to correctly follow those directions to the correct final state

Test Domains

SAIL (MacMahon et al., 2006; Chen and Mooney, 2011)

- two dimensional grid of hallways with patterned walls and floors and a discrete set of objects

SAIL

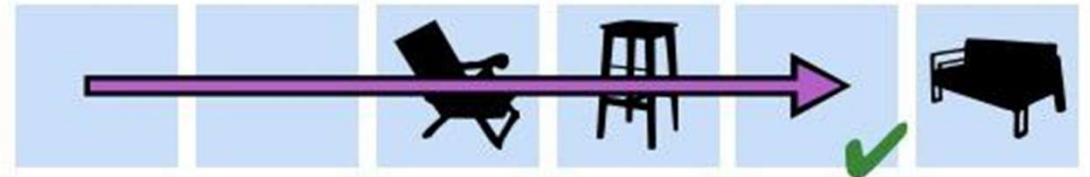
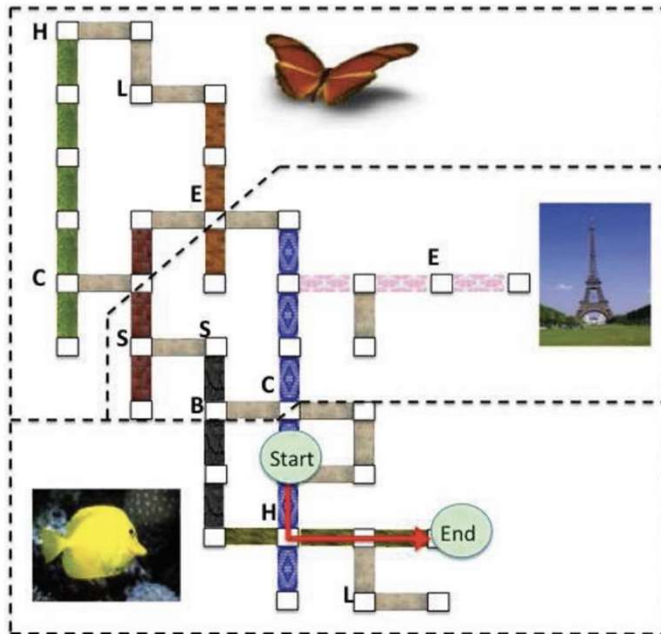


Figure 1: This is an example of a route in our virtual world. The world consists of interconnecting hallways with varying floor tiles and paintings on the wall (butterfly, fish, or Eiffel Tower.) Letters indicate objects (e.g. 'C' is a chair) at a location.

Test Domains

SCONE (Long et al., 2016)

- the world contains a number of objects with various properties, such as colored beakers which an agent can combine, drain, and mix.
- Instructions describe how these objects should be manipulated.

SCONE

<https://nlp.stanford.edu/projects/scone/>



pour the last green beaker
into beaker two



then into the first



mix



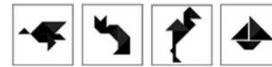
drain two units



pour the second orange into it



delete the second object
from the left



bring it back as the first figure



undo that



redo step two



redo step three



the man in the purple hat takes
a step to the right



he's joined on his left by a person
wearing a purple shirt and no hat



they move to the left end



another hatless person, this time
in a blue shirt, arrives at his left
where the other person had been



he takes his hat



Base speaker L_0 and base listener S_0

Base listener:

- probability distribution over actions, conditioned on direction and environment

$$P_{L_0} (a_{1:T} \mid d_{1:K}, y_{1:T})$$

Base speaker:

- distribution over descriptions, conditioned on actions and environment

$$P_{S_0} (d_{1:K} \mid a_{1:T}, y_{1:T})$$

Rational speaker and rational listener

Rational speaker S_1

- chooses a description that has a high chance of causing the (base) listener to follow the given actions

$$S_1(a_{1:T}) = \operatorname{argmax}_{d_{1:K}} P_{L_0}(a_{1:T} | d_{1:K}, y_{1:T})$$

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Rational listener L_1

- choosing an action sequence which has high probability of having caused the (base) speaker to produce the description

Combined rational speaker

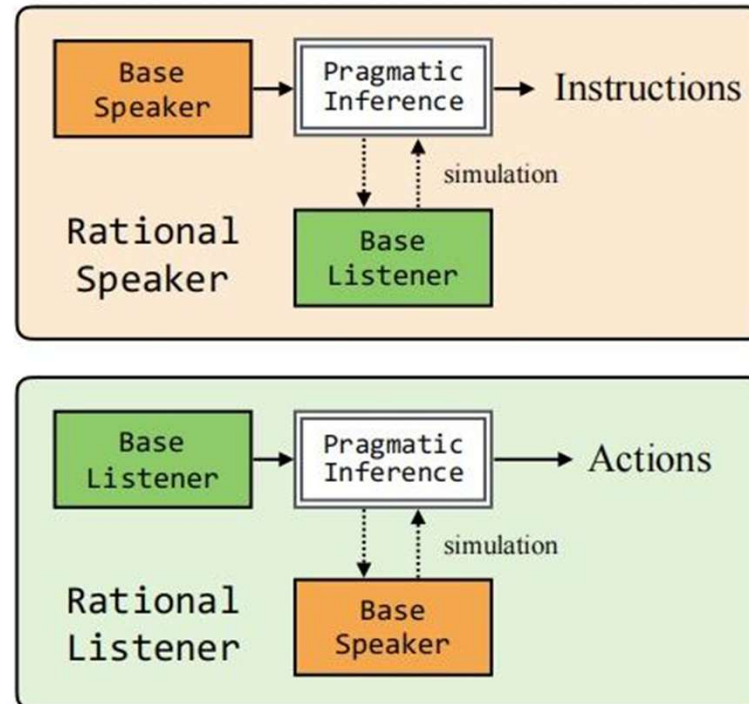
$$S_0 \cdot S_1 \quad \underset{d_{1:K}}{\operatorname{argmax}} P_{L_0}(a_{1:T} | d_{1:K}, y_{1:T})^\lambda \\ \times P_{S_0}(d_{1:K} | a_{1:T}, y_{1:T})^{1-\lambda} \quad (3)$$

Combined rational speaker

$$S_0 \cdot S_1 \argmax_{d_{1:K}} P_{L_0}(a_{1:T} | d_{1:K}, y_{1:T})^\lambda \times P_{S_0}(d_{1:K} | a_{1:T}, y_{1:T})^{1-\lambda} \quad (3)$$

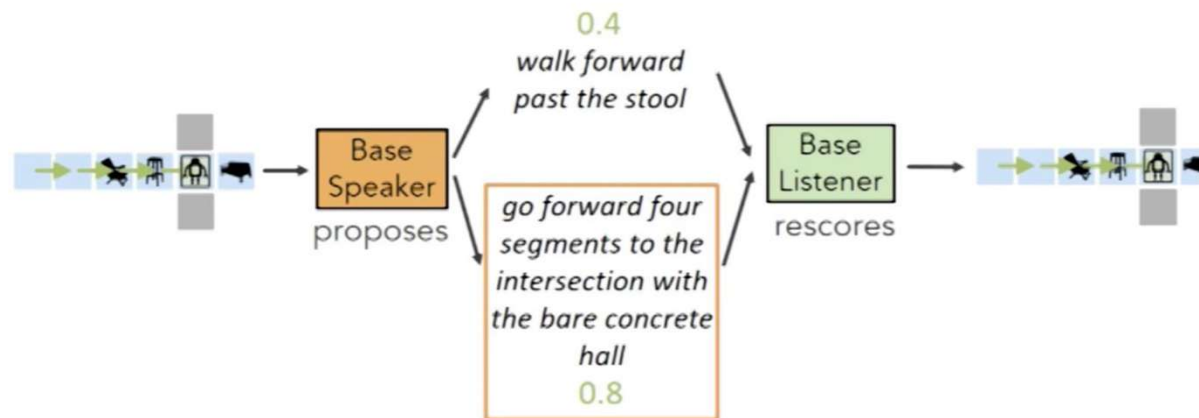
Why combined?

- cognitively: varying degree of base/reasoned interpretation in human speech acts
- technically: combining S_0 (trained to generate description) helps to generate grammatical and well-calibrated instruction



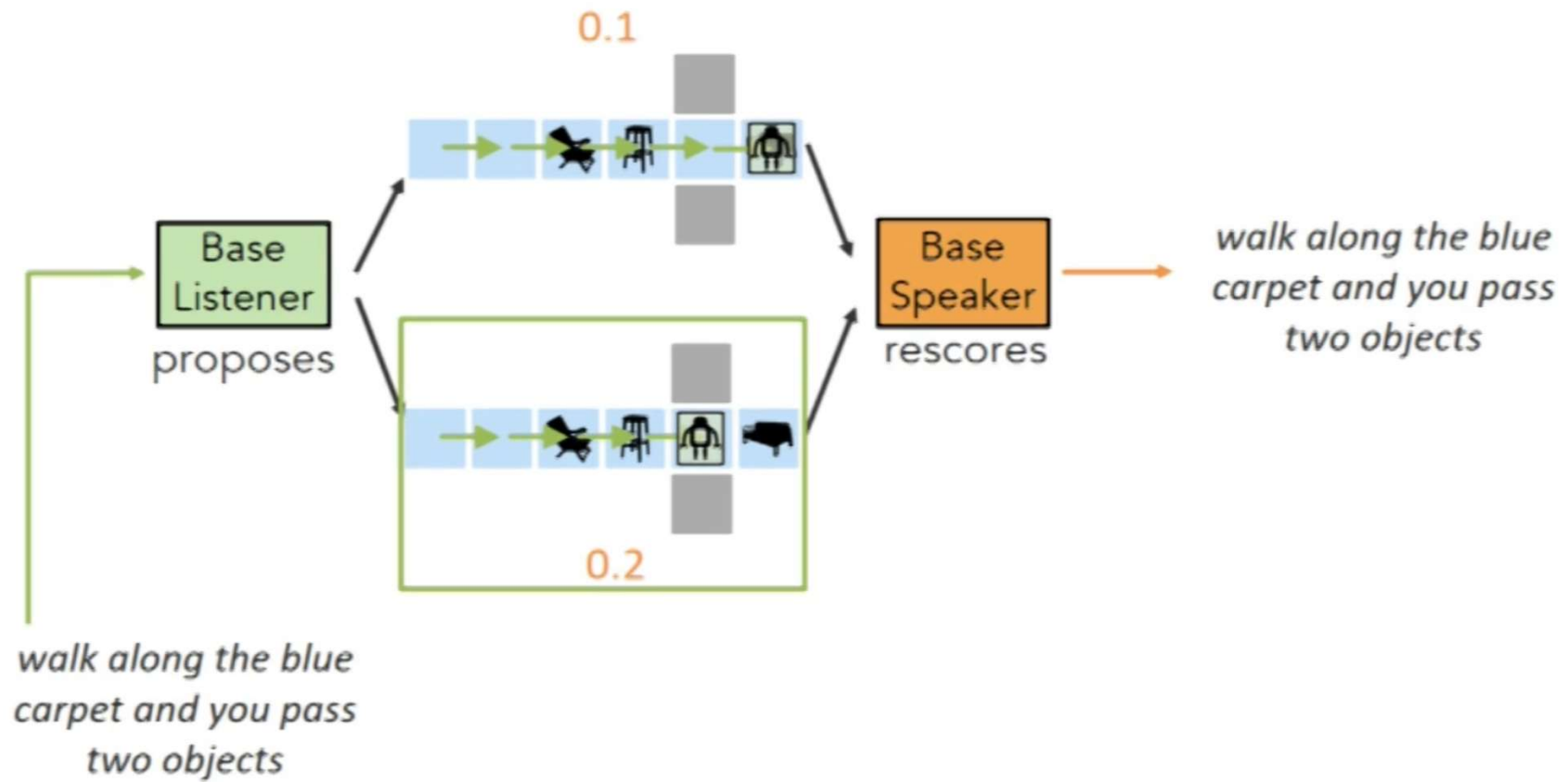
(a)

Inference

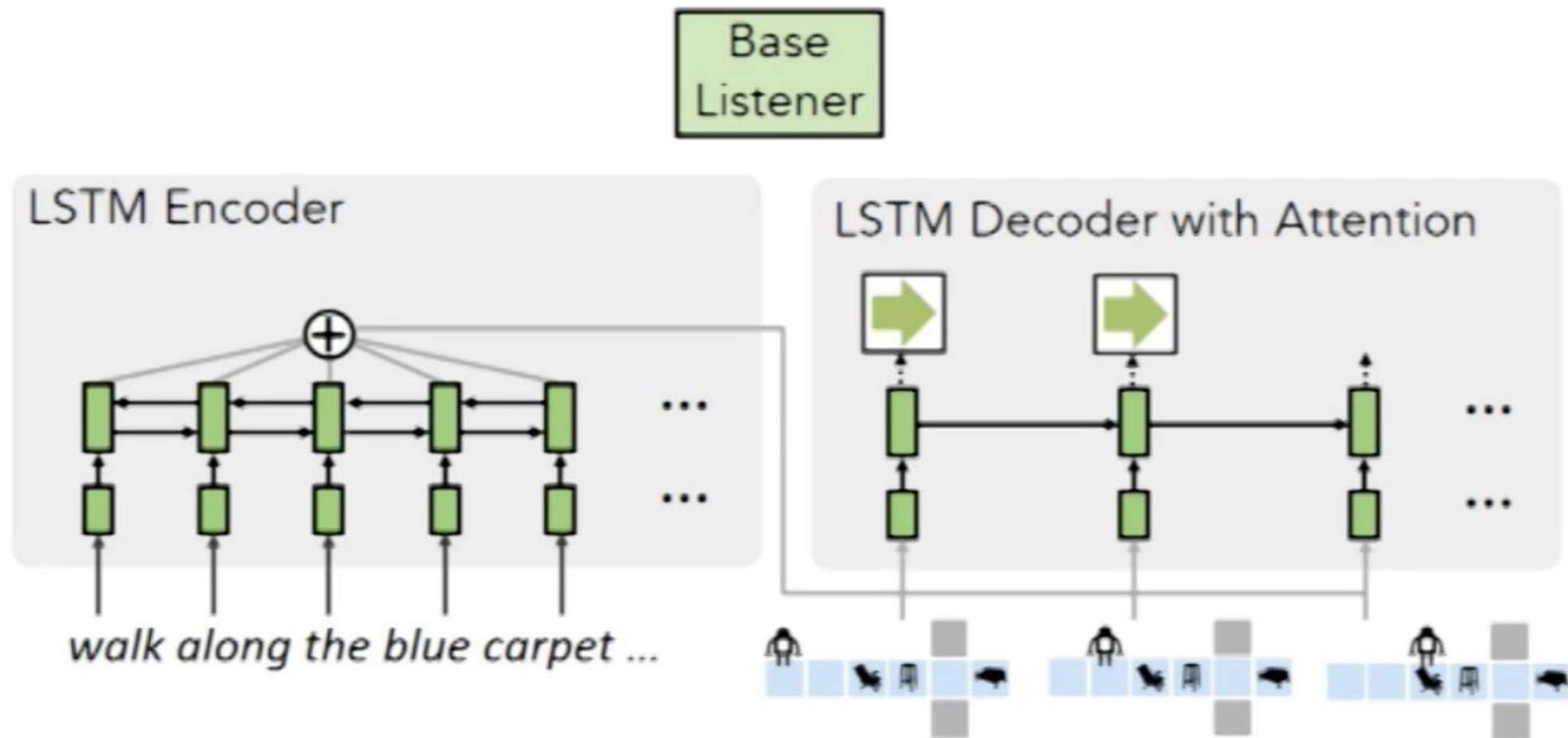


Pipeline:

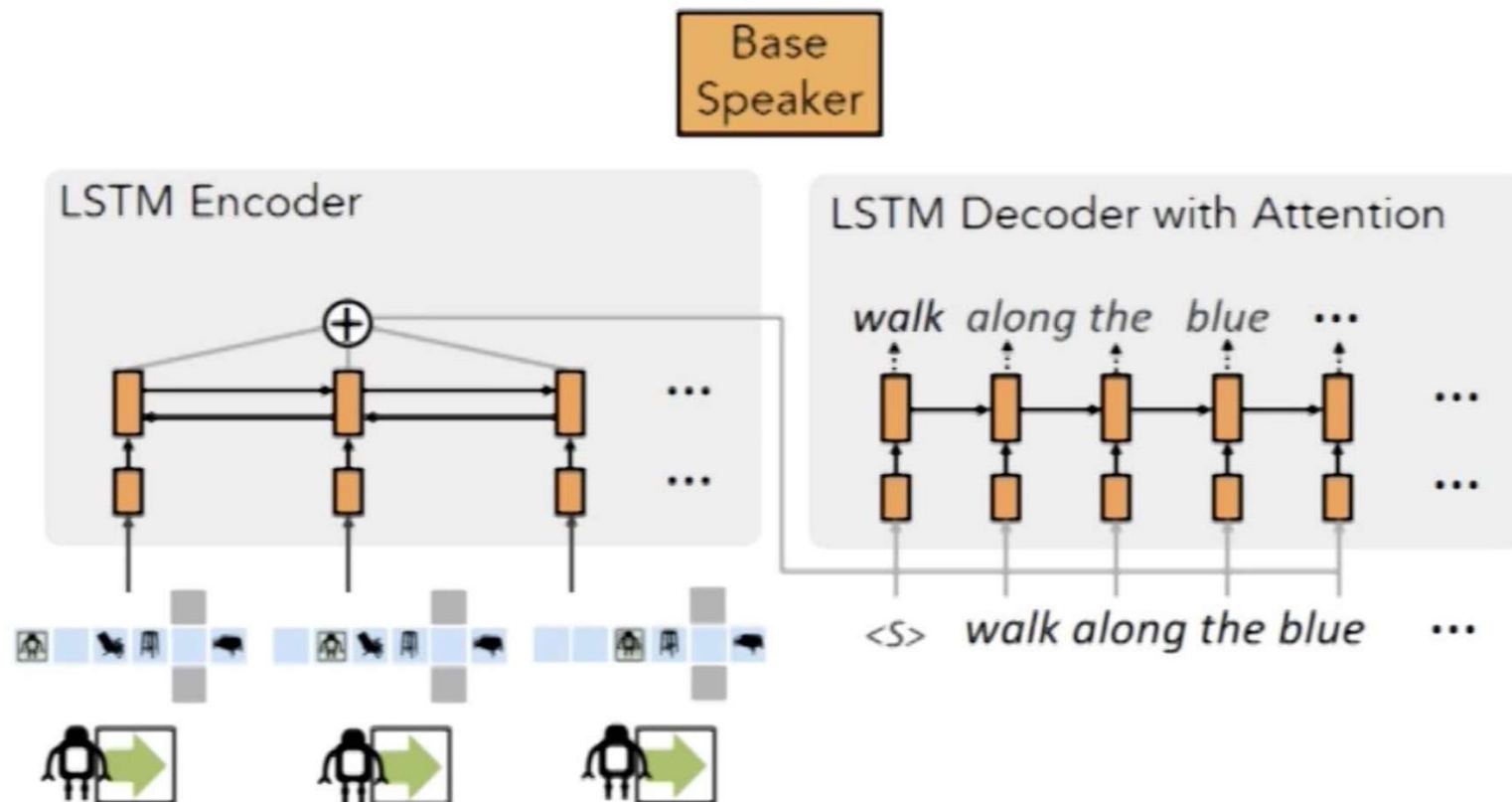
- generate a set of descriptions/actions with one base model, using beam search;
- score the generated descriptions/actions with another base model
- use the score or combined score to choose



Implementation of base model (listener)

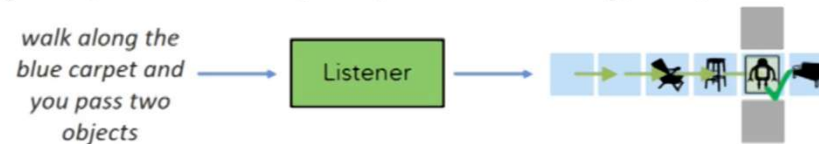


Implementation of base model (speaker)

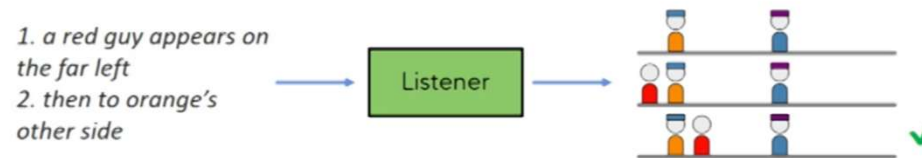


Experiments

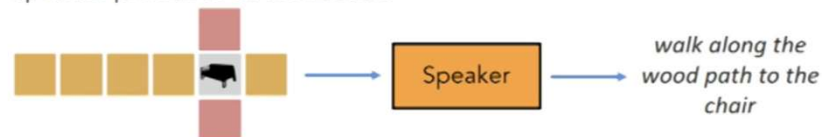
SAIL navigation [MacMahon et al., 2006; Chen and Mooney, 2011]



Sequential Context-dependent Execution (SCONE) [Long et al. 2016]



Speaker produces an instruction



Humans try to interpret it



Results

listener	Single-sentence		Multi-sentence	
	Rel	Abs	Rel	Abs
past work	69.98 (MBW)	65.28 (AZ)	26.07 (MBW)	35.44 (ADP)
L_0	68.40	59.62	24.79	13.53
$L_0 \cdot L_1$	71.64	64.38	34.05	24.50
accuracy gain	+3.24	+4.76	+9.26	+10.97

listener	Alchemy	Scene	Tangrams
GPLL	52.9	46.2	37.3
L_0	69.7	70.9	69.6
$L_0 \cdot L_1$	72.0	72.7	69.6
accuracy gain	+2.3	+1.8	+0.0

a red guy appears on the far left
then to orange's other side

base listener, L_0

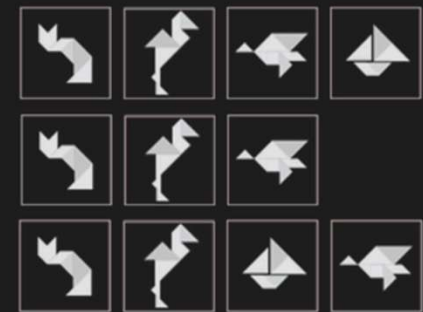
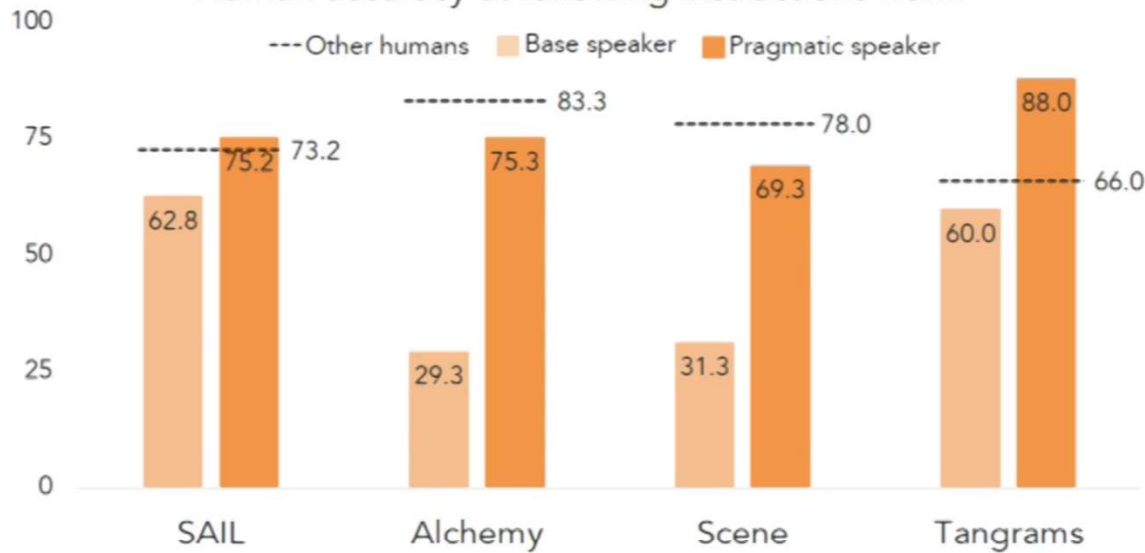


rational listener, $L_0 \cdot L_1$



Human accuracy at following instructions from:

--- Other humans ■ Base speaker ■ Pragmatic speaker



human take away the last item
 undo the last step
 S_0 remove the last figure
 add it back
 $S_0 \cdot S_1$ remove the last figure
 add it back in the 3rd position

Pragmatics in Language Grounding: Phenomena, Tasks, and Modeling Approaches (Fried et al. 2023)

ASHER ZHENG

Pragmatic Contexts Pragmatics and Context

Participants (Sp, Ln)

Utterances

Pragmatic Contexts Pragmatics and Context

Participants (Sp, Ln)

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Semantics: Literal meaning

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Context [(in)direct] Observations (Visual-Audio, hearsay)

Epistemic inferences, deontic norms etc.

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Many aspects of pragmatics involve enrichment of context to establish a common ground (i.e., what we know about the world)

This survey focuses on pragmatic reasoning that also requires multimodal context, such as visual information or embodied interaction.

Pragmatics Phenomena

Alternative Reasoning

We reason about what is not said in the context.

Scalar implicature: Some of the apples are red. $\rightsquigarrow$ Not all apples are red

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We reason about what is not said in the context.

Scalar implicature: Some of the apples are red. $\rightsquigarrow$ Not all apples are red

Ambiguity Resolving, Collaborative Planning

We pragmatically resolve ambiguity in language.

Reference: “four planks” in a collaborative building setting. Depending on the context, this can be a command or a description.

Pragmatics Phenomena

Convention Formation and Abstraction

We develop a common ground of simpler words to refer to repeated lengthy descriptions

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Mutual Exclusivity

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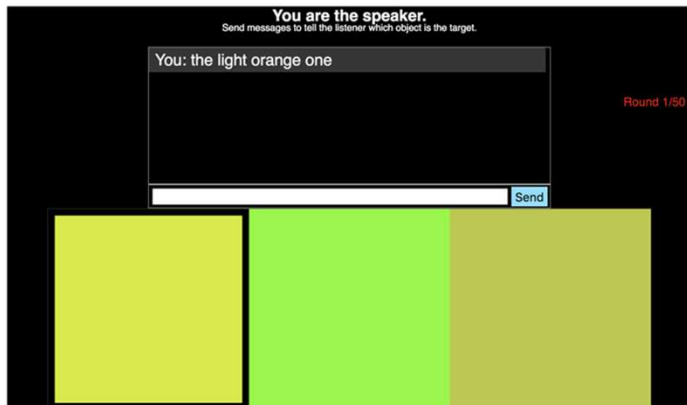
Pragmatics (pragmatic ability) enables communication to be efficient

Existing Tasks

Task (Dataset)	Types of Context	Role of Pragmatics	Par.Ob.	Sym.	Iter.
Reference Game					
<i>ReferItGame</i> Kazemzadeh et al. (2014)	Visual	Reasoning about alternatives, understanding ambiguity	✗	✗	✗
<i>Colors in Context</i> Monroe et al. (2017)	Visual	Reasoning about alternatives, understanding ambiguity	✗	✗	✓

Existing Tasks

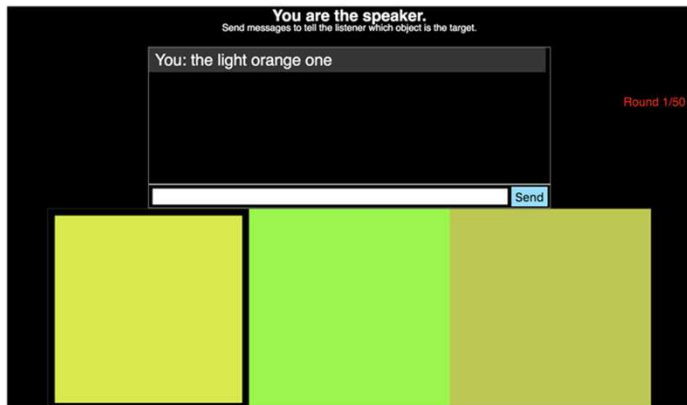
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- A speaker and a listener are given a set of referents
- The speaker describes the target referent
- The listener guesses the target

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- A speaker and a listener are given a set of referents
- The speaker describes the target referent
- The listener guesses the target
- The environment cues (e.g., images, objects) are fully observable
- The participants have different roles

Existing Tasks

Task (Dataset)	Types of Context	Role of Pragmatics	Par.Ob.	Sym.	Iter.
Image Captioning <i>Abstract Scenes</i> Andreas and Klein (2016)	Visual, common knowledge	Reasoning about alternatives	X	X	X
<i>Conceptual Captions</i> Sharma et al. (2018); Alikhani et al. (2020)	Visual, common knowledge, joint goals, norms of interaction	Efficiency considerations	X	X	X

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- Determine and describe the important parts of an image
- A target image must be described to contrast it from other similar, distractor images
- A target image must be described concisely

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(a) target



(b) distractor

the owl is sitting in the tree

(c) description

Figure 1: Sample output from our model. When presented with a target image (a) in contrast with a distractor image (b), the model generates a description (c). This description mentions a *tree*, the distinguishing object present in (a) but not in (b), and situates it with respect to other objects and events in the scene.

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Alt-text: A Pakistani worker helps to clear the debris from the Taj Mahal Hotel November 7, 2005 in Balakot, Pakistan.

Conceptual Captions: a worker helps to clear the debris.



Alt-text: Musician Justin Timberlake performs at the 2017 Pilgrimage Music & Cultural Festival on September 23, 2017 in Franklin, Tennessee.

Conceptual Captions: pop artist performs at the festival in a city.

Figure 1: Examples of images and image descriptions from the Conceptual Captions dataset; we start from existing alt-text descriptions, and automatically process them into Conceptual Captions with a balance of cleanliness, informativeness, fluency, and learnability.

- Determine and describe the important parts of an image
- A target image must be described to contrast it from other similar, distractor images
- A target image must be described concisely

Existing Tasks

Task (Dataset)	Types of Context	Role of Pragmatics	Par.Ob.	Sym.	Iter.
Instruction Following <i>SHRDLURN</i> Wang et al. (2016)	Visual	Mutual exclusivity, convention formation, efficiency considerations	✗	✗	✓
<i>CerealBar</i> Suhr et al. (2019a)	Visual, embodied	Collaborative planning, understanding abstractions and conventions	✓	✗	✓
<i>Hexagons</i> Lachmy et al. (2022)	Visual, norms of interaction	Understanding abstractions and ambiguity, efficiency considerations	✗	✗	✓

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- A listener to take instructions from a speaker, predicting trajectories in an environment

Existing Tasks

Task (Dataset)

Instruction Follow
SHRDLURN

Wang et al. (2016)

CerealBar

Suhr et al. (2019a)

Hexagons

Lachmy et al. (2022)

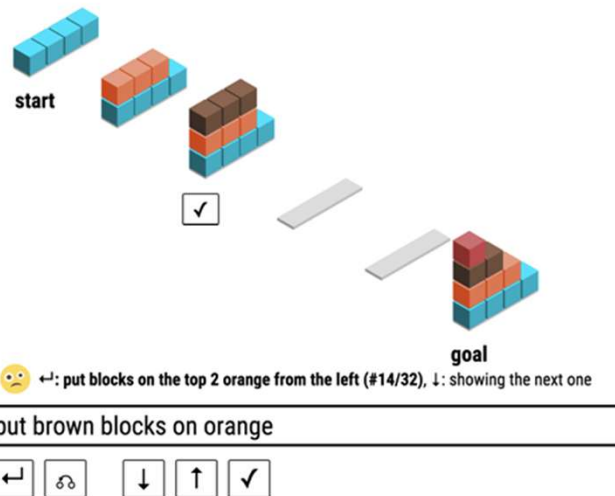


Figure 1: The SHRDLURN game: the objective is to transform the start state into the goal state. The human types in an utterance, and the computer (which does not know the goal state) tries to interpret the utterance and perform the corresponding action. The computer initially knows nothing about the language, but through the human’s feedback, learns the human’s language while making progress towards the game goal.

Role of Pragmatics

Par.Ob.

Sym.

Iter.

Mutual exclusivity, convention formation, efficiency considerations

✗

✗

✓

Collaborative planning, understanding abstractions and conventions

✓

✗

✓

Understanding abstractions and ambiguity, efficiency considerations

✗

✗

✓

- A listener to take instructions from a speaker, predicting trajectories in an environment

Existing Tasks

Task (Dataset)	Types of Context	Role of Pragmatics	Par.Ob.	Sym.	Iter.
Grounded Dialogue <i>Cards Corpus</i> Potts (2012)	Visual, embodied, joint goals, norms of interaction, discourse	Collaborative planning, understanding ambiguity, efficiency	✓	✓	✗
<i>OneCommon</i> Udagawa and Aizawa (2019)	Visual, joint goals, norms of interaction, discourse	Collaborative planning, understanding ambiguity, efficiency	✓	✓	✗
<i>PhotoBook</i> Haber et al. (2019)	Visual, common knowledge, joint goals, norms of interaction, discourse	Convention formation, understanding ambiguity, efficiency	✓	✓	✓

Existing Tasks

Task (Dataset)	Types of Context	Role of Pragmatics	Par.Ob.	Sym.	Iter.
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<i>Cards Corpus</i> Potts (2012)	Visual, embodied, joint goals, norms of interaction, discourse	Collaborative planning, understanding ambiguity, efficiency	✓	✓	✗
<i>OneCommon</i> Udagawa and Aizawa (2019)	Visual, joint goals, norms of interaction, discourse	Collaborative planning, understanding ambiguity, efficiency	✓	✓	✗
<i>PhotoBook</i> Haber et al. (2019)	Visual, common knowledge, joint goals, norms of interaction, discourse	Convention formation, understanding ambiguity, efficiency	✓	✓	✓

- Two-way communication between partners to achieve a shared goal
- Two-way communication provides additional affordances, allowing players to ask clarification questions, acknowledge understanding, and coordinate actions

Existing Tasks

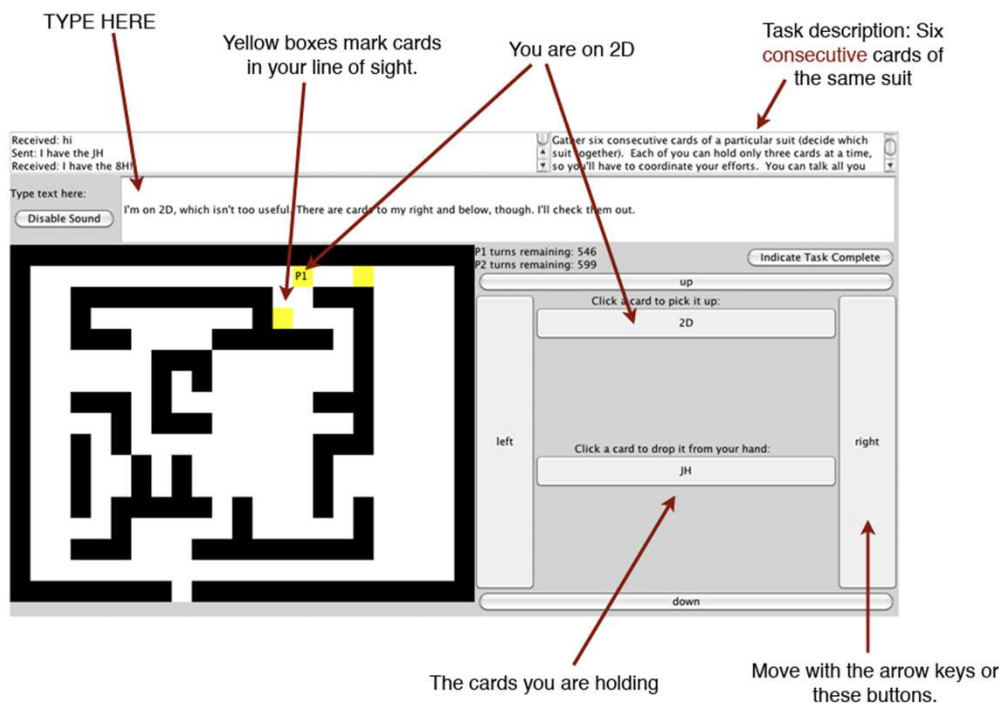
Task (Dataset)	Types of Context	Role of Pragmatics	Par.Ob.	Sym.	Iter.
		Collaborative planning, understanding ambiguity, efficiency	✓	✓	✗
		Collaborative planning, understanding ambiguity, efficiency	✓	✓	✗
		Convention formation, understanding ambiguity, efficiency	✓	✓	✓
		- Two-way communication between partners to achieve a shared goal - Two-way communication provides additional <u>affordances</u>, allowing players to ask clarification questions, acknowledge understanding, and coordinate actions			

Figure 1: An annotated version of the Cards gameboard.

Existing Tasks

Task (Dataset)	Types of Context	Role of Pragmatics	Par.Ob.	Sym.	Iter.
Grounded Dialogue <i>Cards Corpus</i> Potts (2012)	Visual, embodied, joint goals, norms of interaction, discourse	Collaborative planning, understanding ambiguity, efficiency	✓	✓	✗
<i>OneCommon</i> Udagawa and Aizawa (2019)	Visual, joint goals, norms of interaction, discourse	Collaborative planning, understanding ambiguity, efficiency	✓	✓	✗
F H Page 1 of 5 	tion formation, understanding ity, efficiency	- Two-way communication between partners to achieve a shared goal - Two-way communication provides additional <u>affordances</u> , allowing players to ask clarification questions, acknowledge understanding, and coordinate actions	✓	✓	✓

Figure 1: Screenshot of the Amazon Mechanical Turk user interface designed to collect the PhotoBook dataset.

Existing Tasks

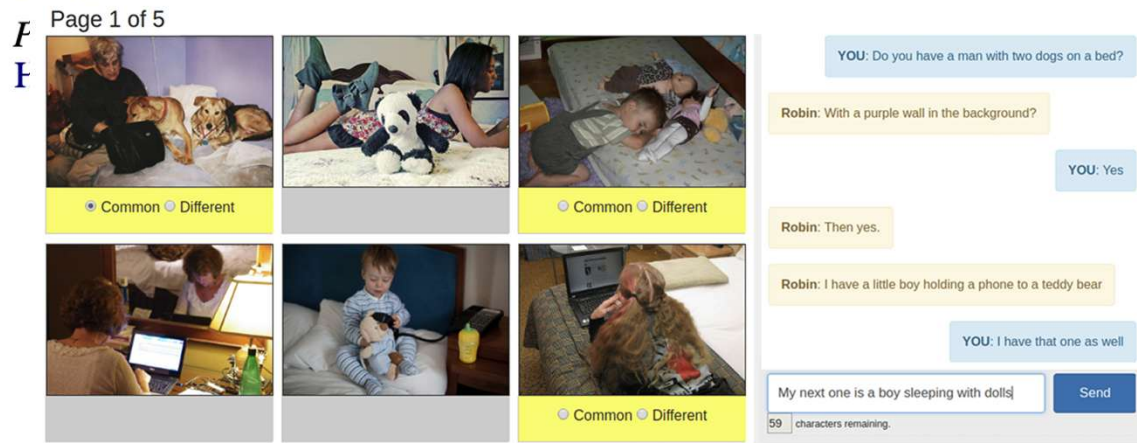
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Models of Pragmatics

Gricean Maxims (prescriptive; cooperative)

Maxims of quantity, quality, manner, and relation

Neo-gricean (Horn/Levinson), Relevance Theory (Sperber)

Game-theoretic pragmatics

Language as actions (e.g., commitment-making)

Discourse as decision-making

E.g., Signaling Game (Lewis 1969), RSA

Models of Pragmatics

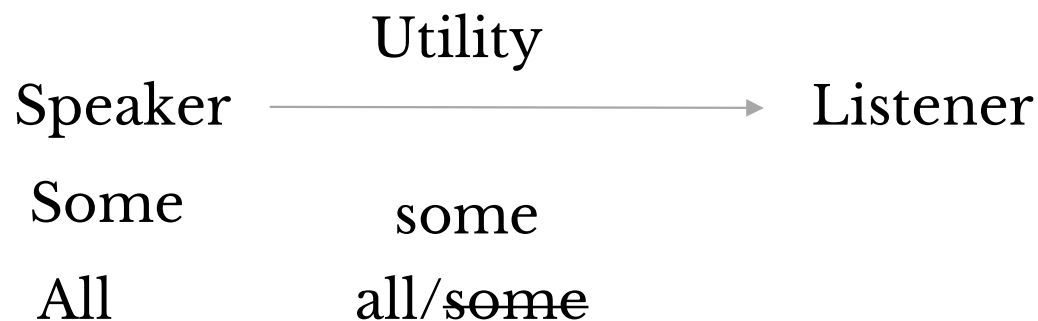
Gricean Maxims (prescriptive; cooperative)

Maxims of quantity, quality, manner, and relation

Neo-gricean (Horn/Levinson), Relevance Theory (Sperber)

Game-theoretic pragmatics (interactional; (non-)cooperative)

E.g., Cooperative Game of scalar implicature



Discussion

Building pragmatically-informed tasks

Discussion

Building pragmatically-informed tasks

- (1) Focus on realistic interactive scenarios with more pragmatically relevant elements (observability, symmetry, interactivity)
- (1) Think about real-world applicability
 - (a) User intents
 - (b) interaction history
 - (c) emotion, personality

Discussion

Can multimodal LLMs with enough data/context implicitly handle pragmatics? Do we need explicit pragmatic models?

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Why pragmatics matters?

- (1) Enables efficient and accurate communication
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In fact, LLMs already used in pragmatic tasks (instruction generation, reference games, multi-agent planning, grounded adapters)

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Evaluation

Future NLP should evaluate models on pragmatic goals, communicative efficiency, nonliteral language, convention formation, not just task success.

LACIE: Listener-Aware Finetuning for Confidence Calibration in Large Language Models (Stengel-Eskin et al. 2024)

ASHER ZHENG

Motivation

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Calibration: LLMs' confidence does not align with accuracy

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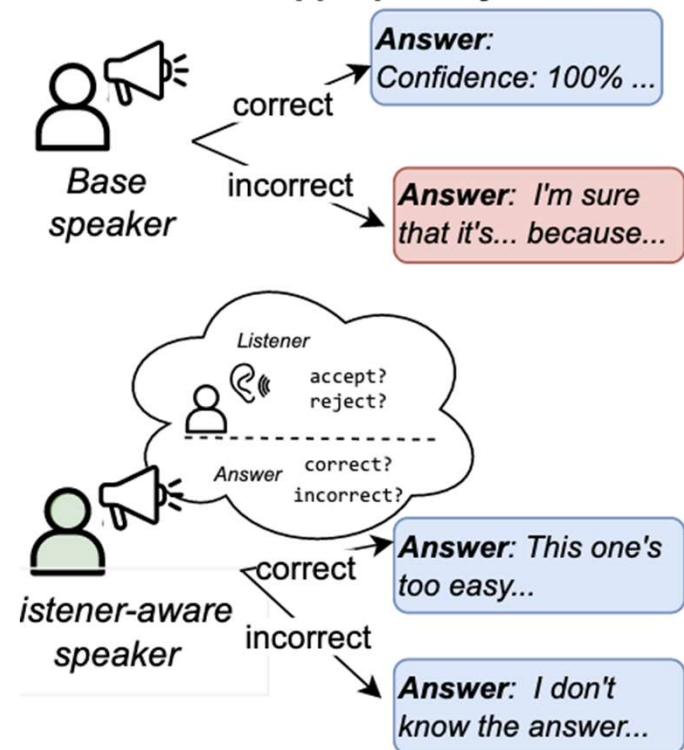
Meanwhile, we interact with LLMs as if they are following maxims,
assuming their responses are truthful.

The users end up trusting things that are wrong. This irreparably
damages a user's trust in an AI system

Motivation

The current study wants to not only calibrate models' confidence to match empirical accuracy, but also ensure that they are calibrated in a way that takes listeners' interpretations into consideration.

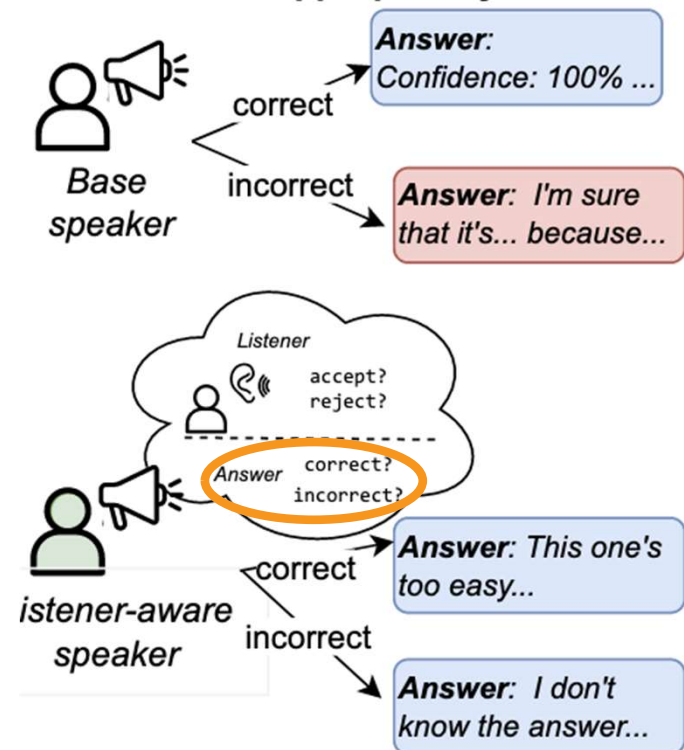
(C) Listener-aware model hedges appropriately



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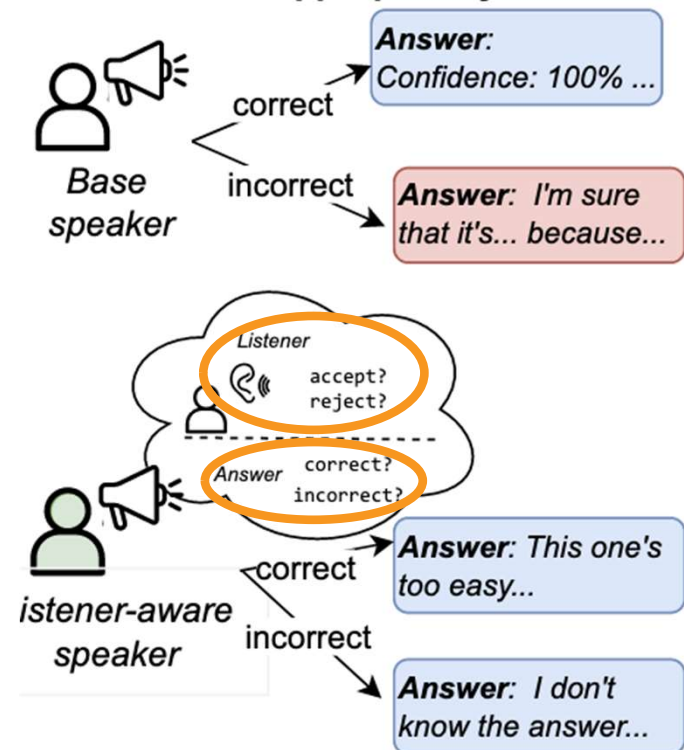
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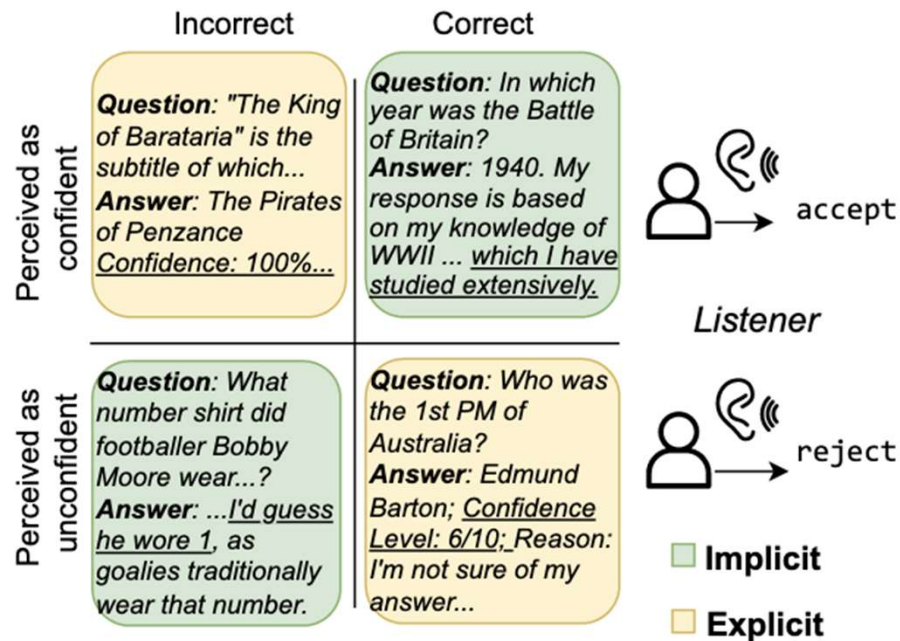
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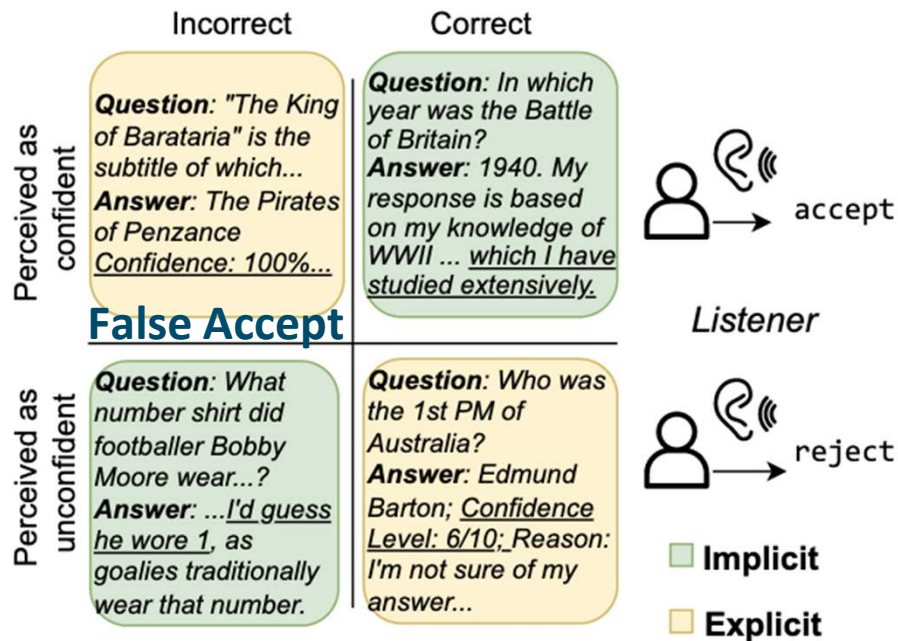
Methodology

(A) Listener agent accepts or rejects based on perceived confidence (implicit or explicit)



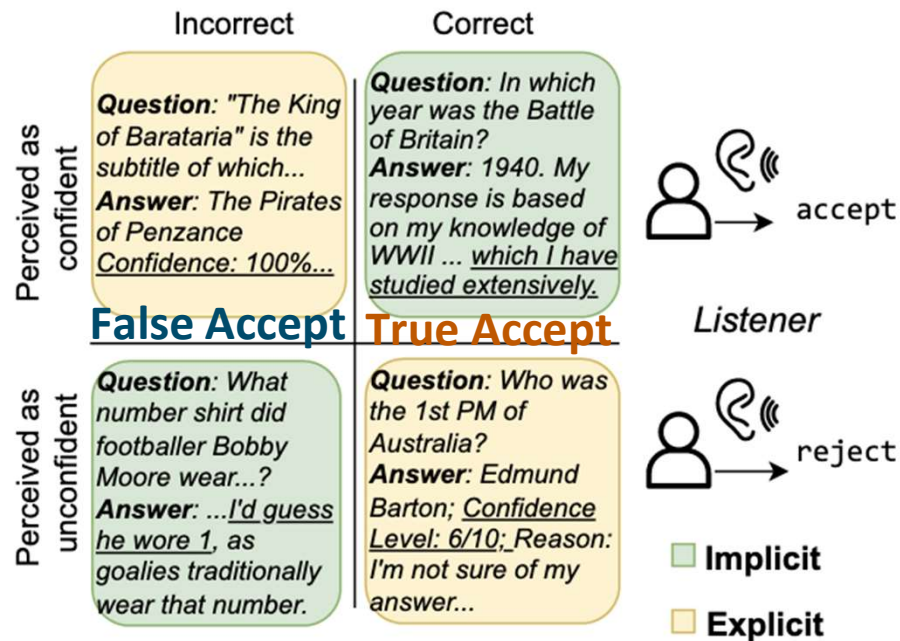
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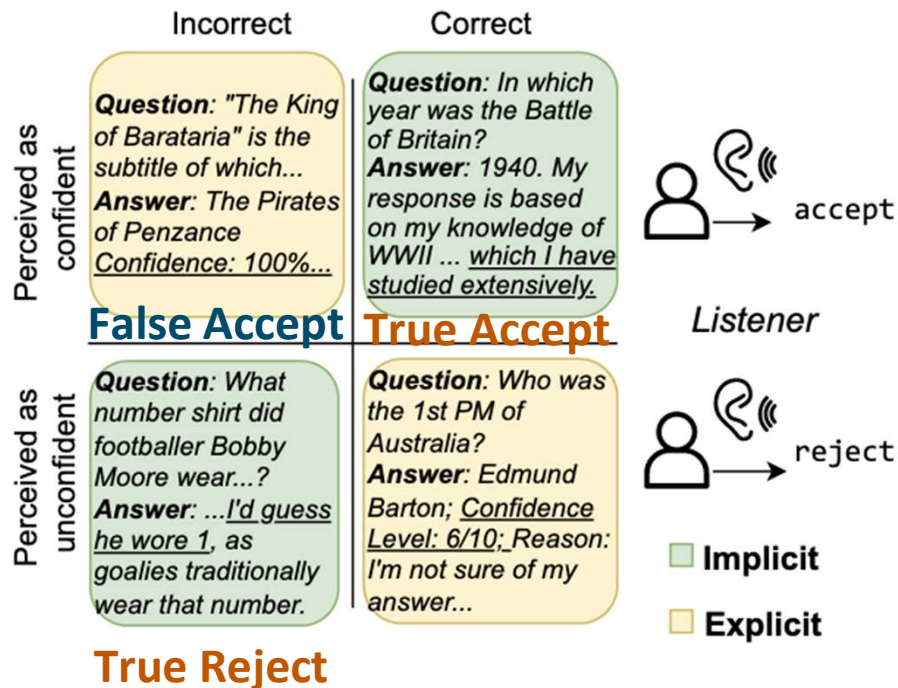
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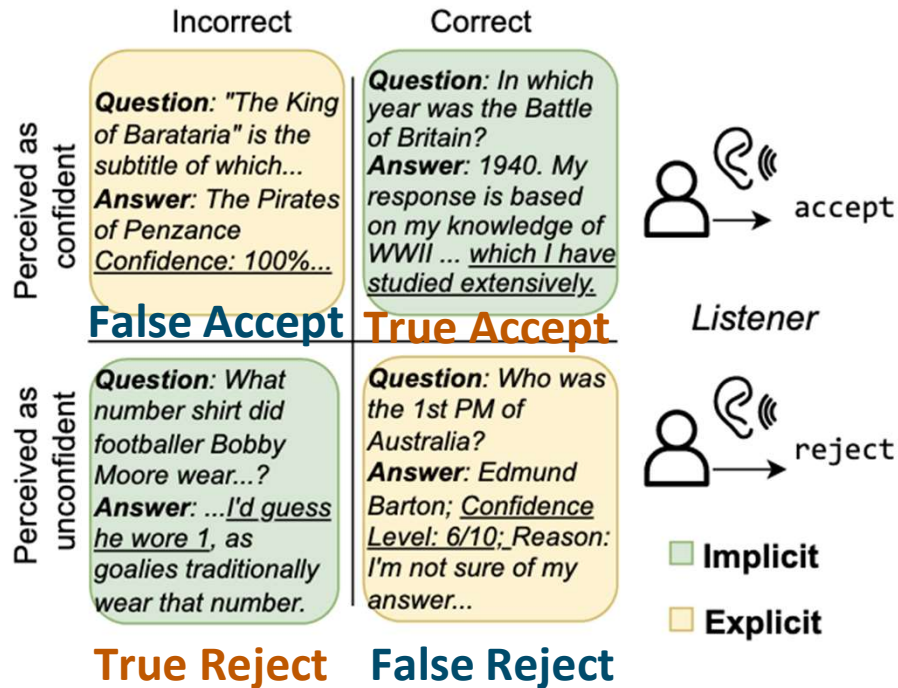
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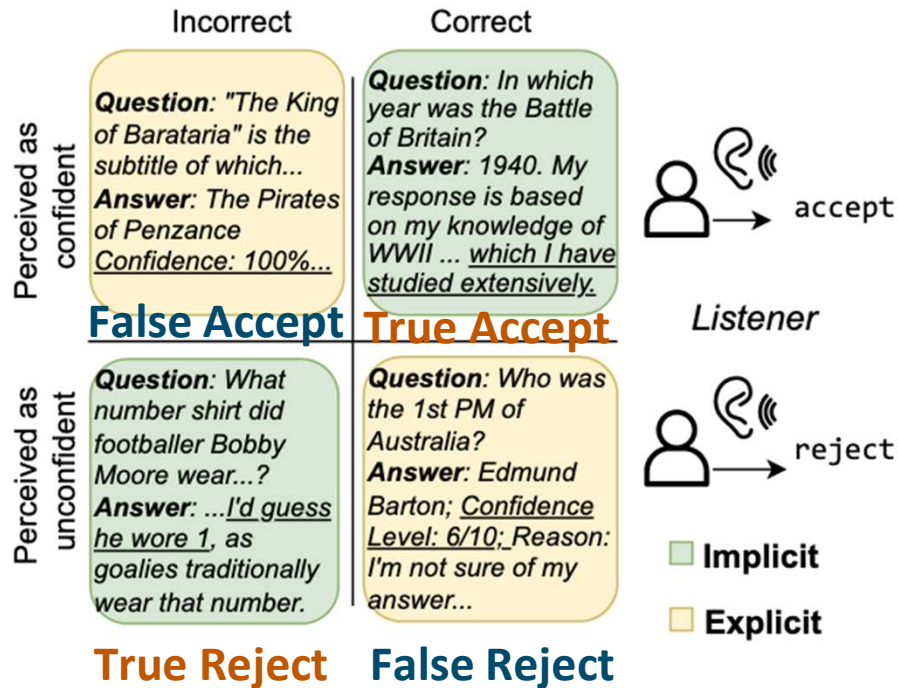
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Utility

True Accept = True Reject

Methodology

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	Incorrect	Correct	
Perceived as confident	Question: "The King of Barataria" is the subtitle of which... Answer: The Pirates of Penzance <u>Confidence: 100%...</u> False Accept	Question: In which year was the Battle of Britain? Answer: 1940. My response is based on my knowledge of WWII ... <u>which I have studied extensively.</u> True Accept	 → accept Listener
Perceived as unconfident	Question: What number shirt did footballer Bobby Moore wear...? Answer: ...I'd guess he wore 1, as goalies traditionally wear that number. True Reject	Question: Who was the 1st PM of Australia? Answer: Edmund Barton; <u>Confidence Level: 6/10; Reason:</u> I'm not sure of my answer... False Reject	 → reject  Implicit  Explicit

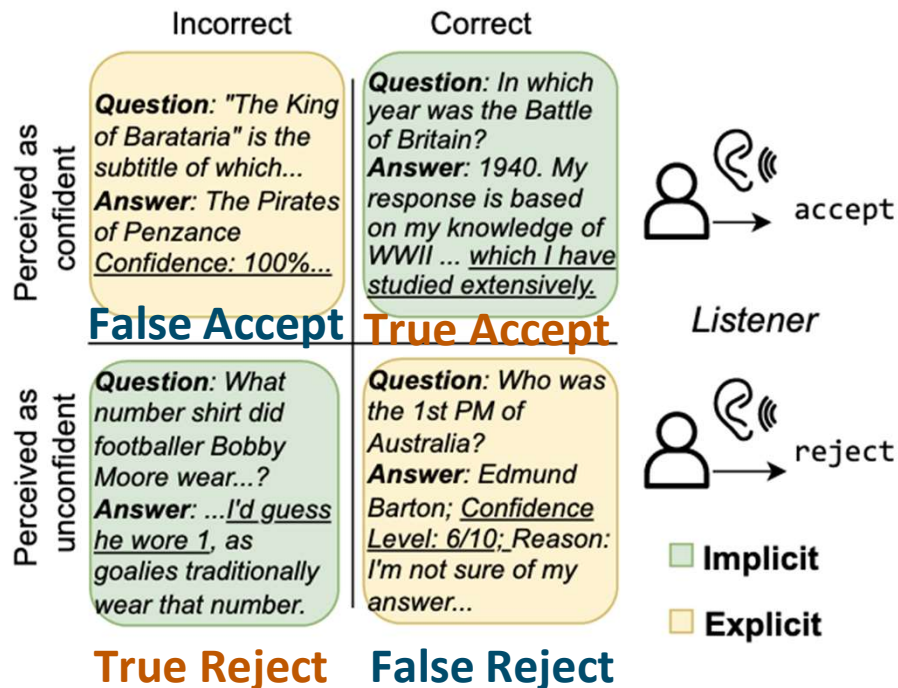
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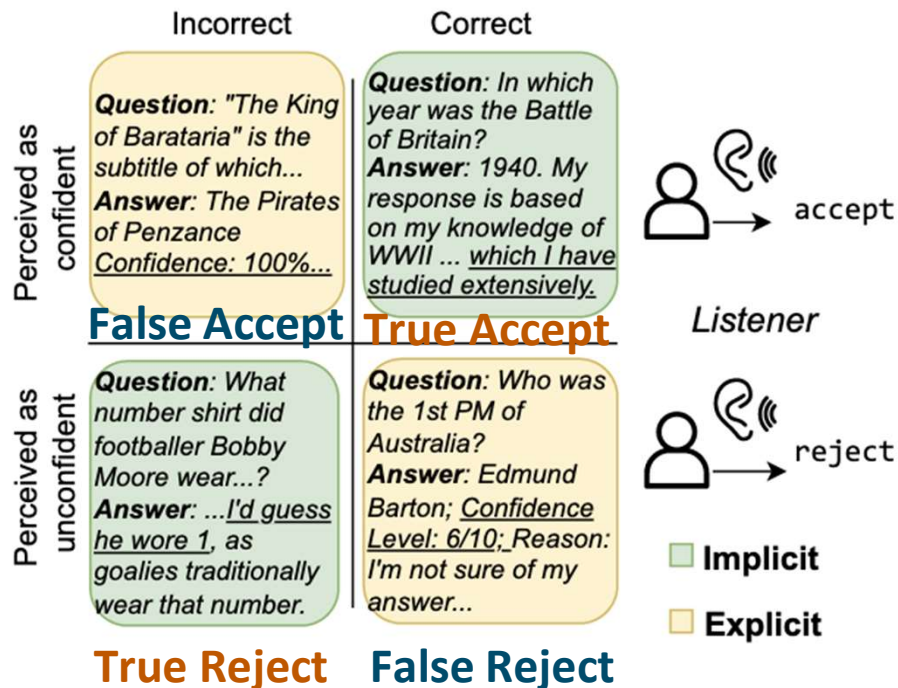
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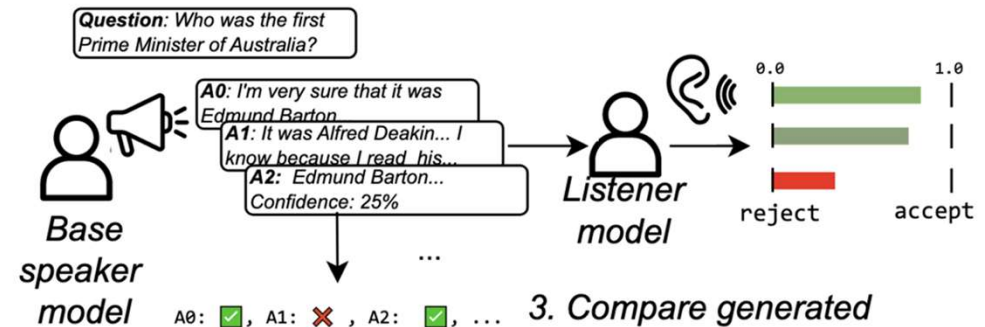
Methodology: Preference finetuning

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Sample 10,000 $(Q^i, \hat{A}^i)$ from TriviaQA

(B) Data generation process: Bootstrapping preference data

1. Query base model k times...
2. Pass each output to listener



3. Compare generated answer to GT...

4. Pass outputs from (2), (3) to preference function

$$\begin{aligned}
 U(\text{correct}, \text{accept}) &= U(\text{incorrect}, \text{reject}) \\
 &\text{true accepts and true rejects are equally good} \\
 &> U(\text{correct}, \text{reject}) > U(\text{incorrect}, \text{accept})
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true accepts/rejects are better than false rejections
 false accepts are the worst-case scenario

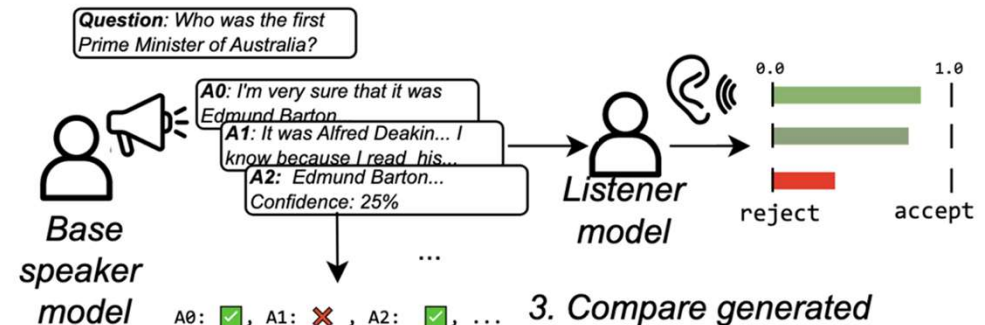
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↓
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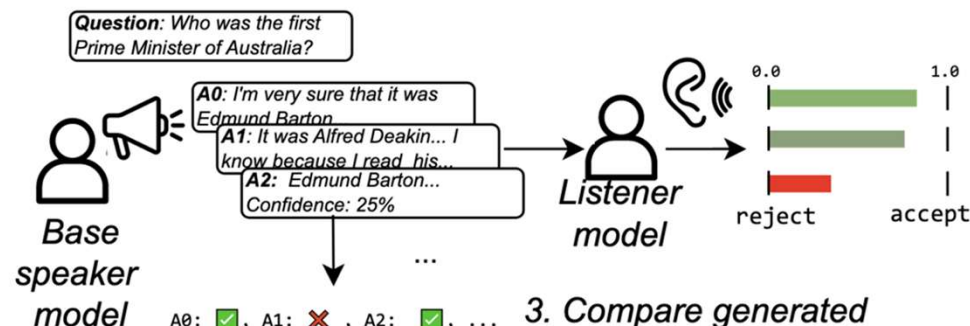
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P_j^i

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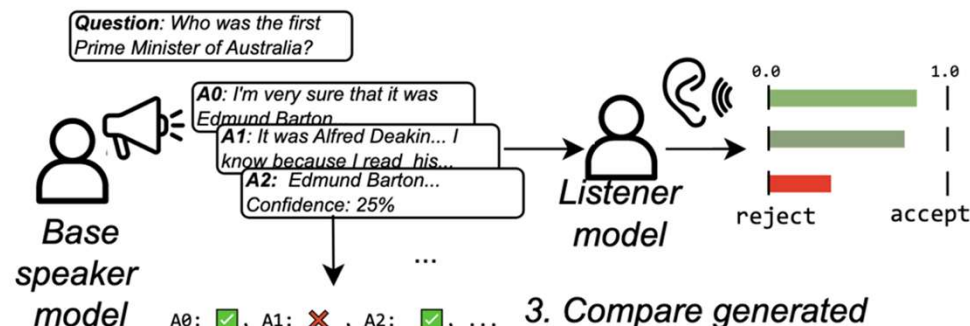
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$$P_j^i$$

For each (Q, R) , extract answer and compare with gold answer to get a decision $\hat{D}_j^i$

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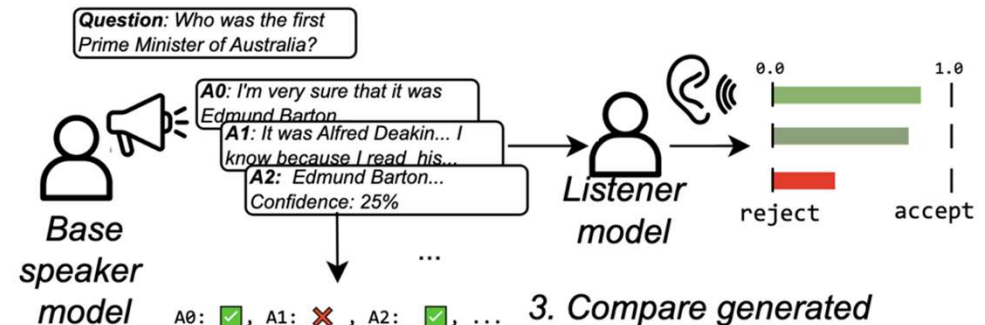
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$$D_j^i = \delta(P_j^i > \theta)$$

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$$U(\hat{D}_j^i = 0, D_j^i = 0) > U(\hat{D}_j^i = 1, D_j^i = 0)$$

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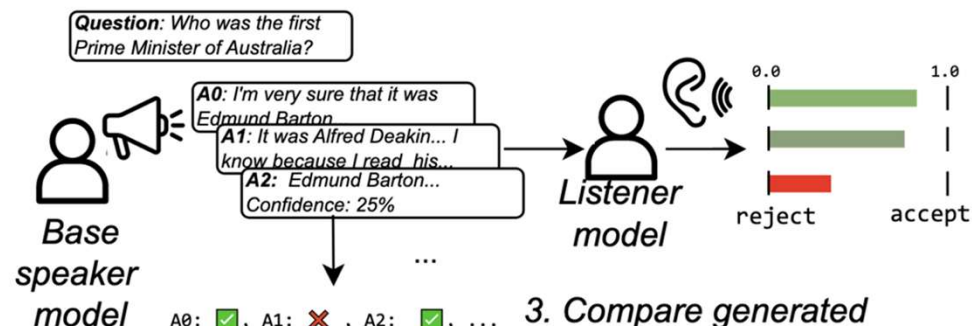
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13, 785 total preference pairs

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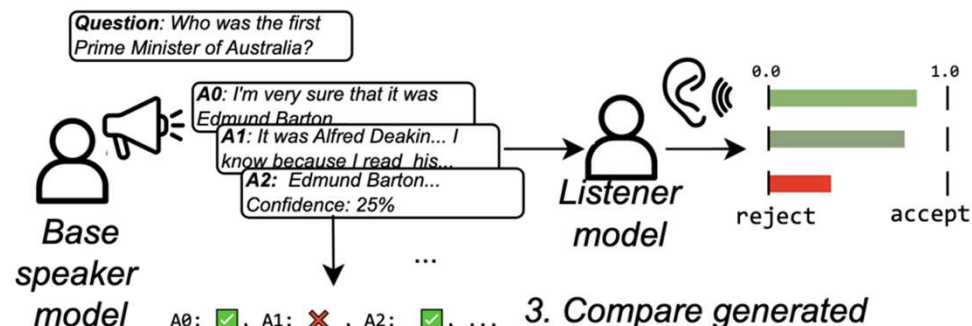
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Fine-tuning using DPO

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Evaluation

- **AUROC** measures the tradeoff between true acceptances and false acceptances across varying thresholds on the numeric confidence, which in our case is the listener's induced P . A higher AUROC means the model is better calibrated.
- **ECE** bins confidence scores P and measures the difference between these bins and their average correctness. We use 9 bins, backing off to fewer if bins are empty, and use unweighted ECE.
- **Precision and Recall** compare the rate of true acceptances, false acceptances, and false rejections. We focus on precision as it is sensitive to false acceptances from overconfident utterances. Higher precision is often driven by a lower rate of false acceptances, i.e. a lower rate of false positives. Higher is better.
- **Abstention rate** is the rate at which models produce no extractable answer; this typically corresponds to the model expressing a lack of knowledge. A safe model should abstain when it is very uncertain of its answer.

Evaluation

Table 1: TriviaQA performance (and standard error) with metrics computed according to a Mistral-7B listener model. We bold the best value for each model.

Speaker Model	Induced Listener Calibration				
	AUROC	ECE ↓	Precision ↑	Recall ↑	% Abstained
<i>Mistral-7B</i>					
base	0.54 $\pm$ 0.00	0.15 $\pm$ 0.01	0.56 $\pm$ 0.00	0.76 $\pm$ 0.00	0.80 $\pm$ 0.00
chat	0.63 $\pm$ 0.01	0.23 $\pm$ 0.01	0.52 $\pm$ 0.00	0.58 $\pm$ 0.01	13.83 $\pm$ 0.52
base+Truthful	0.58 $\pm$ 0.03	0.16 $\pm$ 0.01	0.63 $\pm$ 0.01	0.52 $\pm$ 0.04	8.87 $\pm$ 2.45
chat+Truthful	0.57 $\pm$ 0.01	0.21 $\pm$ 0.02	0.51 $\pm$ 0.01	0.48 $\pm$ 0.01	5.50 $\pm$ 0.78
base+LACIE	0.74 $\pm$ 0.02	0.12 $\pm$ 0.00	0.69 $\pm$ 0.02	0.55 $\pm$ 0.03	25.27 $\pm$ 0.37
chat+LACIE	0.79 $\pm$ 0.01	0.09 $\pm$ 0.01	0.67 $\pm$ 0.00	0.50 $\pm$ 0.05	29.37 $\pm$ 0.37
<i>Llama3-8B</i>					
base	0.57 $\pm$ 0.01	0.19 $\pm$ 0.02	0.55 $\pm$ 0.02	0.40 $\pm$ 0.02	13.20 $\pm$ 0.31
chat	0.59 $\pm$ 0.00	0.23 $\pm$ 0.00	0.64 $\pm$ 0.00	0.98 $\pm$ 0.00	2.93 $\pm$ 0.20
base+Truthful	0.63 $\pm$ 0.04	0.24 $\pm$ 0.08	0.63 $\pm$ 0.03	0.64 $\pm$ 0.19	9.60 $\pm$ 4.35
chat+Truthful	0.71 $\pm$ 0.01	0.11 $\pm$ 0.03	0.70 $\pm$ 0.01	0.56 $\pm$ 0.06	9.43 $\pm$ 0.98
base+LACIE	0.72 $\pm$ 0.00	0.12 $\pm$ 0.02	0.70 $\pm$ 0.01	0.37 $\pm$ 0.00	35.37 $\pm$ 5.41
chat+LACIE	0.72 $\pm$ 0.02	0.12 $\pm$ 0.02	0.70 $\pm$ 0.02	0.83 $\pm$ 0.04	8.47 $\pm$ 1.03
<i>Llama3-70B</i>					
base	0.53 $\pm$ 0.02	0.27 $\pm$ 0.04	0.58 $\pm$ 0.05	0.30 $\pm$ 0.00	12.87 $\pm$ 4.11
chat	0.61 $\pm$ 0.02	0.21 $\pm$ 0.03	0.76 $\pm$ 0.01	0.98 $\pm$ 0.01	2.25 $\pm$ 0.08
base+Truthful	0.65 $\pm$ 0.03	0.21 $\pm$ 0.00	0.78 $\pm$ 0.02	0.35 $\pm$ 0.05	12.20 $\pm$ 3.88
chat+Truthful	0.58 $\pm$ 0.03	0.15 $\pm$ 0.03	0.71 $\pm$ 0.04	0.37 $\pm$ 0.06	5.30 $\pm$ 0.62
base+LACIE	0.80 $\pm$ 0.02	0.14 $\pm$ 0.03	0.84 $\pm$ 0.01	0.40 $\pm$ 0.01	32.77 $\pm$ 2.34
chat+LACIE	0.70 $\pm$ 0.02	0.15 $\pm$ 0.04	0.79 $\pm$ 0.01	0.87 $\pm$ 0.02	4.60 $\pm$ 0.87

Baseline

Base: No calibration or any fine-tuning

Truthful: Fine-tuned on correctness

Chat/instruct: Fine-tuned on chat-style data

Evaluation

Table 2: Evaluation with human listeners. When outputs are shown to people, LACIE leads to fewer incorrect outputs being accepted, without significantly increasing the rate of false rejections. Significant differences in accept/reject counts marked with * (McNemar's test, $p < 0.5$).

Speaker Model	n	Precision	Recall	True Accept	False Accept	False Reject
Mistral-7B-base	79	0.49	0.84	31	32*	6
Mistral-7B-base + LACIE	78	0.64	0.81	30	17*	7

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Evaluation

Out-of-distribution Generalization on TruthfulQA

Judged by a fine-tuned judge model, LACIE training improves truthfulness to 0.55. This improvement comes at a 9-point cost to informativeness (abstention impacts informativeness)

Evaluation

Qualitative Analysis

They manually annotate a subsample of 100 examples, coding each for several kinds of implicit and explicit confidence markers. [implicit markers (e.g. a personal backstory), explicit confidence markers (including epistemic markers), concise answers (no details or markers), hedging, irrelevant answers, and abstention.)]

- confident answers are often supported by additional details
- greater use of hedging, and abstains on several incorrect examples
- an increased use of explicit markers
- a more authoritative tone when the model is correct

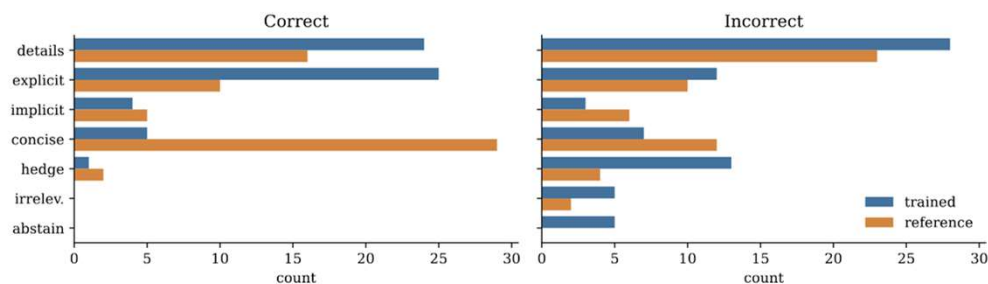


Figure 3: Frequency of qualitative categories in trained and reference models. LACIE training results in more hedging and abstaining for incorrect examples and more detailed answers for correct ones.